QML-IDS Draft Review

Draft vs. Independent Re-Run · UNSW-NB15 + CIC-IDS2017 · Seed 42 · For revision before resubmission

Headline verdict: the draft's core numbers reproduce well. An independent re-run on a larger slice of UNSW-NB15 lands within 1–2 points of every reported metric (accuracy, F1, MCC, ROC-AUC). That's a genuinely strong result to state explicitly in the paper — most QML-IDS papers never show their numbers survive a second run. Below: exact deltas, three things worth double-checking, and ready-to-paste text for the sections this new data fills in.

1. QSNN-NIDS: Draft vs. Re-Run (UNSW-NB15)

The draft draws 50,000 records from only the official training partition and re-splits them (7,284-flow test set). The re-run instead combines the official training and test CSVs (a loader bug had been silently dropping the test file) and re-splits that full 200,000-row set (26,028-flow test set) — a larger, more representative sample, same architecture and seed (42).

Metric	Draft (n=7,284)	Re-run (n=26,028)	Delta
Accuracy	0.8585	0.8630	+0.0045
Precision (attack)	0.7904	0.8369	+0.0465
Recall (attack)	0.8981	0.8903	−0.0078
F1 (attack)	0.8408	0.8628	+0.0220
FPR	0.1698	0.1626	−0.0072
MCC	0.7190	0.7276	+0.0086
ROC-AUC	0.9397	0.9382	−0.0015
Mean spike rate	0.284	0.144	−0.140 (worth checking)
Latency/flow	3.55–3.79 ms	7.91 ms	hardware-dependent, not a bug

Three Things Worth a Closer Look Before Citing Both Runs Together

Spike rate dropped by half (28.4% → 14.4%) despite near-identical detection metrics. Worth a quick check: same `angle_gain` / `angle_bias` /threshold in both configs? If so, this is itself a reportable finding (the model reaches similar accuracy with much sparser firing on more data) rather than a bug — but it should be confirmed, not just footnoted.

Random Forest baseline is noticeably lower in the re-run (93.34% → 90.48% accuracy, FPR 7.17% → 10.76%). Since RF has no stochasticity tied to the quantum encoder, this gap most likely comes from the larger/different test distribution (official test partition included) rather than from anything in the QSNN pipeline. Suggest stating this plainly if both numbers appear in the same table.

Latency more than doubled (3.6–3.8ms → 7.9ms). Expected and fine to report as-is — different machine, different PennyLane build — but the paper should state both are simulator latencies on different CPUs, not imply they're comparable like-for-like.

2. What This Run Adds to the Draft's Own Limitations Section

Section VI of the draft lists six explicit gaps. The re-run closes three of them outright and makes real (if partial) progress on a fourth:

RESOLVED *"Full noisy runs and real-hardware runs were not part of this experiment."*

A 6-point depolarizing-noise sweep (p = 0–0.1) was run on both datasets using PennyLane's noisy density-matrix simulator. Result: the model is robust through p=0.02 (97–100% F1 retention on both datasets), then fails — but differently per dataset. On UNSW-NB15 it collapses to majority-class prediction at p≥0.05 while ROC-AUC stays near 0.91 (the decision threshold breaks, not the underlying signal). On CIC-IDS2017 it degrades more gracefully at p=0.05 (F1=0.28, ROC-AUC still 0.98) before failing completely at p=0.1 (ROC-AUC drops to 0.47, i.e. random). Real IBM hardware was attempted (backend `ibm_fez`) but blocked by the account's exhausted usage quota — worth stating explicitly as "attempted, blocked by provider quota" rather than "not attempted."

RESOLVED *"Did not evaluate on... CIC-IDS2017."*

Full pipeline run end-to-end: 300,000 rows, all 8 release files (one had a latin1-encoding issue, now fixed), QSNN + all 5 baselines + 6-point noise sweep. See table below.

RESOLVED *"Quantum baselines (QSVC, QCNN) were only exercised on synthetic smoke-test data, so they are not reported here."*

Both now have full results on both real datasets (table below) — this directly fulfills future-work item (v) in the draft's own Section VII.

PARTIAL *"Results come from a single seed and a single split."*

Still true — multi-seed repeats were attempted but not completed in the available time (interrupted to prioritize the noise sweep). Recommend keeping this limitation as-is rather than overstating progress, with a note that it's in progress.

3. New Results to Add — CIC-IDS2017

Model	Family	Accuracy	Precision	Recall	F1	FPR	AUC

QSNN-NIDS	Quantum	0.9746	0.9187	0.9560	0.9370	0.0208	0.9939
Random Forest	Classical	0.9965	0.9877	0.9946	0.9912	0.0030	0.9997
RBF-SVM	Classical	0.9901	0.9603	0.9910	0.9754	0.0101	0.9983
MLP (DNN)	Classical	0.9910	0.9668	0.9882	0.9774	0.0084	0.9994
QSVC (ZZ kernel)	Quantum	0.9720	0.9400	0.9216	0.9307	0.0151	0.9717
QCNN	Quantum	0.9715	0.8967	0.9674	0.9307	0.0275	0.9953

Same qualitative story as UNSW-NB15 — classical baselines still ahead, but the gap is narrower in absolute terms, and QSNN-NIDS posts its best numbers of the two datasets here (F1 0.937 vs. 0.863 on UNSW-NB15). Worth a sentence in Discussion: the model's relative ranking vs. classical baselines is consistent across datasets, which is itself a generalization claim the single-dataset draft couldn't make.

4. Suggested Text Edits (Ready to Adapt)

Add — Section V, New Subsection: Noise Robustness

Ready-to-paste text

We evaluated the trained QSNN-NIDS model under a depolarizing-plus-amplitude-damping noise channel at six levels (p = 0, 0.005, 0.01, 0.02, 0.05, 0.1), applied after every variational block, using PennyLane's density-matrix simulator. On UNSW-NB15, the model retains 97–98% of its ideal F1 through p=0.02, then collapses at p≥0.05 to predicting the majority class (F1=0), while ROC-AUC remains at 0.91–0.86 — indicating the underlying decision scores still separate the classes; only the fixed classification threshold fails under heavy noise. On CIC-IDS2017 the model is noise-tolerant through a wider range (99–101% F1 retention through p=0.02, and a graceful degradation to F1=0.28 at p=0.05 with ROC-AUC still 0.98), but fails completely at p=0.1, where ROC-AUC itself drops to 0.47 (chance level) — here the underlying signal degrades, not just the threshold. This dataset-dependent failure mode suggests that noise robustness in quantum IDS models should be characterized per-deployment rather than assumed to transfer.

Add — Section V, New Subsection: Cross-Dataset Evaluation (CIC-IDS2017)

Ready-to-paste text

To test whether QSNN-NIDS generalizes beyond UNSW-NB15, we repeated the full pipeline — identical architecture, hyperparameters and seed — on CIC-IDS2017 (Monday–Friday capture, 300,000 flows sampled across all eight release files, reduced via the same MI-filter-then-PCA pipeline to 8 features). The model reaches 97.46% accuracy, F1 0.937 and ROC-AUC 0.994, outperforming its own UNSW-NB15 numbers on every metric. Classical baselines retain their lead (Random Forest: 99.65% accuracy, F1 0.991), but the margin is narrower than on UNSW-NB15, and the model's ranking relative to every baseline is consistent across both datasets.

Revise — Section VI, Limitations

Replace the three sentences below with the updated text:

Original: "All circuits were simulated without noise… full noisy runs and real-hardware runs were not part of this experiment… We used 50,000 records… and did not evaluate on the official test partition or on other datasets such as CIC-IDS2017… Quantum baselines (a ZZ-kernel QSVC and a QCNN)… were only exercised on synthetic smoke-test data, so they are not reported here."

Suggested replacement

We additionally evaluated the model under simulated depolarizing/amplitude-damping noise (Section V-?) and on a second, independent dataset, CIC-IDS2017 (Section V-?), with full QSVC and QCNN baselines on both. A real-hardware run was attempted on IBM's ibm_fez backend but could not complete because the account's allotted quantum-processing time was exhausted at the time of the experiment; this remains open. Results throughout the paper are still from a single seed and a single split per dataset; multi-seed repeats to report confidence intervals are left for future work.

Revise — Section VII, Future Work

Items (i), (iv) and (v) can be marked done or partially done rather than left as open future work:

- (i) run the model under realistic noise models and on IBM quantum hardware → **DONE** (noise); **ATTEMPTED, BLOCKED BY QUOTA** (hardware)
- (iv) evaluate on CIC-IDS2017 … with repeated seeds → **DONE** (single seed); repeated seeds still open
- (v) complete the comparison with QSVC and QCNN at full scale → **DONE**, both datasets

Remaining future work: (ii) learn the firing threshold and study other encodings, (iii) multi-class detection of individual attack families, (iv-cont.) repeated-seed confidence intervals and ToN-IoT, (vi) neuromorphic deployment for real energy measurement, and completing the real-hardware run once IBM quota is available.

One methodological point to flag to the authors, not just patch over: the draft's 50k-record, training-partition-only sample and the re-run's full 200k-record (official train+test combined) sample are not the same protocol. If both sets of numbers go in the same paper, state this difference explicitly next to the comparison — e.g. as a dedicated "Reproducibility Check" paragraph or table footnote — rather than presenting them as the same experiment at two scales. Reviewers who catch an unstated protocol change read it as a bigger problem than a disclosed one.